

(Internet Engineering Task Force), and was rechristened as HTTP 2.0 after minor modifications. This led to the morphing of the stack, as shown in Figure 5. This was a very important step as the key Web protocol, HTTP, was being overhauled roughly after a decade.

However, Google and others continued their data collection efforts to verify that HTTP 2.0 had indeed helped with the latency problem. The data indicated that HTTP 2.0 was still limited by TCP's slow start and congestion control and, thereby, indicated further scope for optimising latency. Based on this, QUIC was proposed as a potential replacement for TCP, and this is still being discussed at IETF.

Concurrently, the security researchers improved TLS 1.2 to TLS 1.3, bringing in both performance and security gains. That brings us to the present, and Figure 6 represents the stack as of today.

Separately, it was recognised that DNS was one of the last protocols that sent unencrypted data over the network. While this is not the content required by the user, it nevertheless represented metadata that could still be of importance

to attackers. To plug this, a secure version of DNS, named DNS-over-HTTPS was proposed and standardised by IETF.

Overview of the key protocols of the early era

To recap, the key Web protocols of the early era, i.e., during 1990s, are:

1. TCP/IP
2. HTTP
3. DNS
4. SSL

As our focus is not on core networking, we will only cover the HTTP, DNS and SSL protocols briefly. However, there is extensive literature on these protocols on the Web. As there is no great benefit in duplicating any of that, the objective here is to discuss the key attributes of the protocol that will help an interested reader, at least as a starting point, for further study.

DNS: DNS, the Domain Name System, is provided by the collective effort of the following entities:

1. DNS client
 - a. DNS resolver in OS, called the stub resolver.

2. DNS servers

- a. Recursive resolvers – the entities that receive the DNS requests from stub resolvers.
- b. Authoritative DNS servers – which host the various DNS attributes for host names.

The above entities 'talk' the DNS protocol, which is in binary. The client initiates a DNS request, which contains one or more questions. The key attributes of each question are *Hostname* (or fully qualified domain name) and *DNS property type*.

Various DNS property types are:

- A (request for IPv4 address of the accompanying hostname)
- AAAA (request for IPv6 address)
- MX (request for mail server addresses)
- TXT, and so on

The DNS servers parse the question(s) and either forward it (them) or respond. The DNS response contains answers that either contain an error code or the information sought by the client. It also contains metadata, like TTL or the time duration for which the response is valid.

A sample DNS capture from Wireshark examples is given in Figure 7 to help the reader get a feel of the structure of DNS packets. This is an IPv4 address lookup for the hostname *www.netbeans.org*.

The response packet returns the IPv4 address for the requested domain and indicates the TTL for the same. The flags indicate the following:

1. This response was not obtained from the authoritative server. So, it must have been returned from the cache entry of some recursive resolver in the path.
2. The DNS response is not authenticated, which means the DNS server did not employ DNSSEC.

HTTP: The HTTP protocol shares some similarities with the DNS protocol, as it also follows the client-server paradigm. The HTTP client initiates a request which, unlike DNS, is in regular text. Each request has a

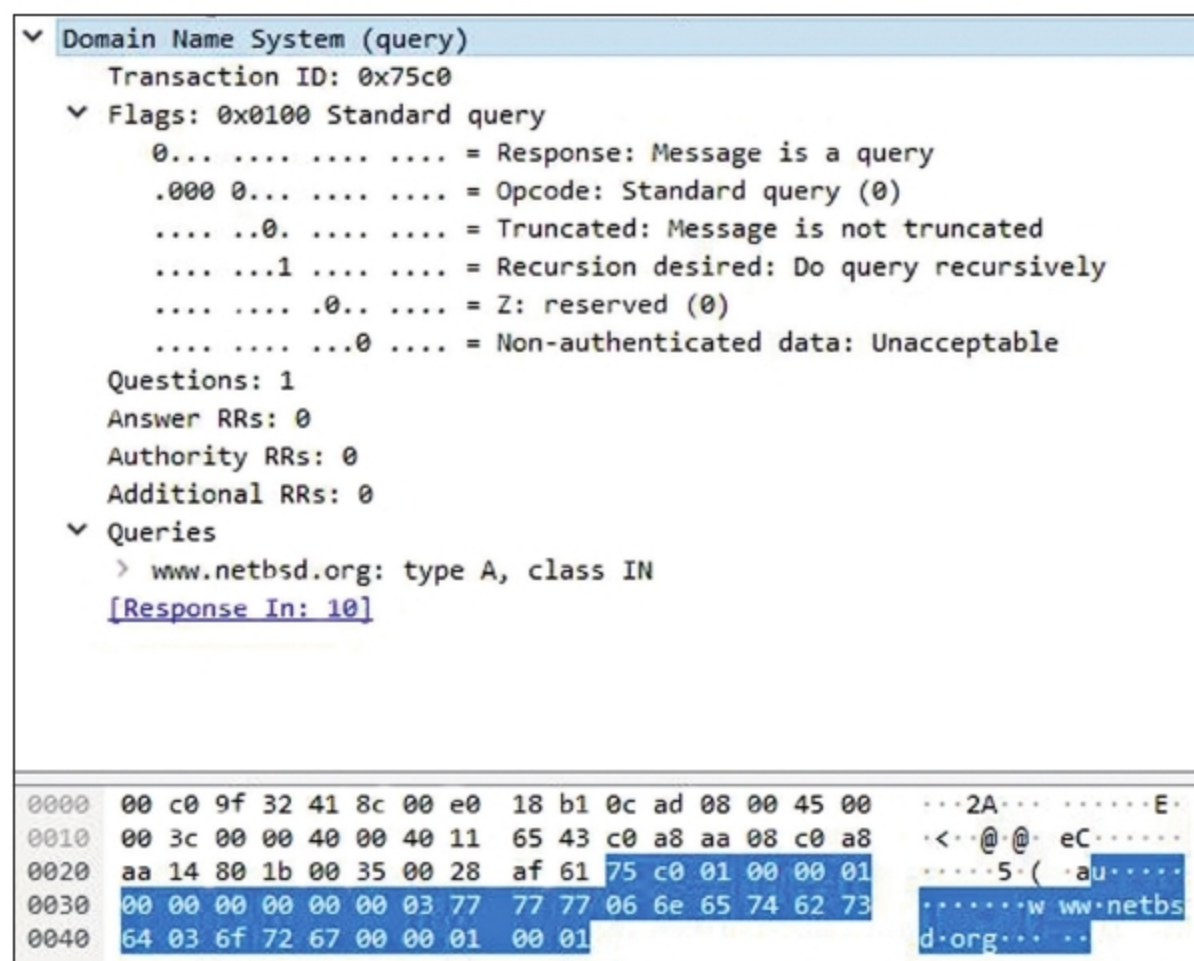


Figure 7: DNS capture from Wireshark